

PREMIER UNIVERSITY

Department of Computer Science & Engineering



Big Data Analytics Lab (CSE 4346- V4)

Project Report

on

“College Major & Career Outcomes Analysis Using Hadoop HDFS,
MapReduce, and Apache Pig”

Submitted by:

Md. Mahafujul Alam

ID: 0222220005101066

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Submitted To:

Md. Tamim Hossain

Lecturer,

Department of Computer Science & Engineering

Premier University, Chittagong

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Remarks:

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1. Problem Statement

The rapid growth of educational and career-related data has created large datasets containing information about college majors, institutions, earnings, employment, student debt, occupations, and career outcomes. Processing and analyzing such large-scale datasets using traditional data-processing methods can be challenging due to their size and diverse structure.

This project analyzes the College Major & Career Outcomes 2026 dataset using Hadoop HDFS, Java MapReduce, and Apache Pig. The dataset contains 227,981 records and 72 attributes, with a raw size of approximately 126.34 MB.

The main analyses performed in this project are:

1. Average 4-year earnings by college major.
2. Total employment by major category.
3. Average student debt by major category.
4. Average earnings by major category using Apache Pig.
5. Total employment by major category using Apache Pig.
6. Identification of the top 10 major categories by employment.
7. Identification of programs with high earnings and relatively low student debt.

2. Dataset Description

2.1 Source

The dataset used in this project is the College Major & Career Outcomes 2026 dataset, collected from Kaggle. The dataset contains information about college programs, educational institutions, earnings, employment outcomes, student debt, occupations, and AI-related workforce indicators.

Dataset Link: [College_Major_Career_Analytics_2026](#)

The HDFS input location used in the project is:

/BDAFinal1066/college/college_major_career_outcomes_2026.csv

2.2 Dataset Statistics

The dataset contains 227,981 records and 72 attributes, with a raw file size of approximately 126.34 MB.

Property	Details
Dataset Name	College Major & Career Outcomes 2026
Source	Kaggle
File Format	CSV
Total Records	227,981
Total Attributes	72
File Size	126,340,696 bytes
Approximate Size	126.34 MB
Distinct Major Titles	358
Major Categories	44

The dataset contains a combination of categorical, numerical, textual, percentage, geographic, and status-related fields, making it suitable for Big Data processing and analytical operations.

2.3 Dataset Fields

The dataset contains 72 attributes covering different aspects of college programs and career outcomes. The major field groups are:

Category	Example Fields
Institution	Institution name, state, region, control
Academic Program	CIP code, CIP title, credential level
Earnings	1-year, 4-year, and 5-year median earnings
Employment	Working count, working percentage, employment outcomes
Student Debt	Median debt, monthly payment, debt-to-earnings ratio
Occupation	Linked occupations, occupation wage, growth
AI & Technology	AI software share, AI tools, technology indicators

Important Fields Used in Analysis

Field	Description
cip_title	College major title
cip_family_title	Major category
median_earnings_4yr_usd	Median earnings after four years
median_earnings_1yr_usd	Median earnings after one year
median_earnings_5yr_usd	Median earnings after five years
count_working_in_state_5yr	Number working in-state after five years
median_debt_usd	Median student debt
median_monthly_payment_usd	Median monthly debt payment
earnings_growth_pct_1yr_to_5yr	Earnings growth from one to five years
occupation_median_wage_2024_usd	Median occupation wage
occupation_growth_pct_2024_34	Occupation growth percentage
ai_software_occupation_share	Share of occupations using AI software

3. Summary of Data Analysis

The College Major & Career Outcomes 2026 dataset was analyzed using Hadoop HDFS, Java MapReduce, and Apache Pig. The analysis focuses on college major earnings, employment, and student debt.

3.1 Analytical Operations

The project includes three MapReduce operations:

1. Average Earnings by Major – calculates the average 4-year earnings for each major.
2. Total Employment by Category – calculates total working graduates for each major category.

3. Average Debt by Category – calculates average student debt for each major category.

The project also includes five Apache Pig operations:

1. College Major Analysis
2. Average Earnings by Category
3. Total Employment by Category
4. Top 10 Employment Categories
5. High Earnings and Low Debt Programs

3.2 Workflow Diagram

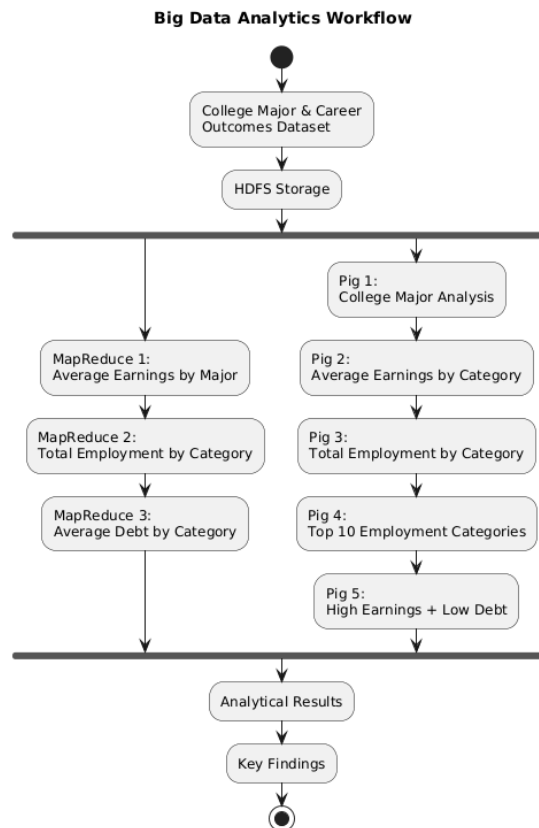


Fig 1: Big Data Analytics Workflow

The diagram illustrates the complete processing workflow, from storing the dataset in HDFS to performing three MapReduce and five Pig operations. The generated outputs are used to produce the final analytical results and key findings.

4. HDFS Analysis

Hadoop Distributed File System (HDFS) was used as the storage layer for the College Major & Career Outcomes 2026 dataset. The dataset was uploaded to HDFS and verified before performing MapReduce and Pig operations.

4.1 HDFS Directory Structure

A project directory was created in HDFS: /BDAFinal1066/college

The command used was: `hdfs dfs -mkdir -p /BDAFinal1066/college`

The directory was used to store the project dataset.

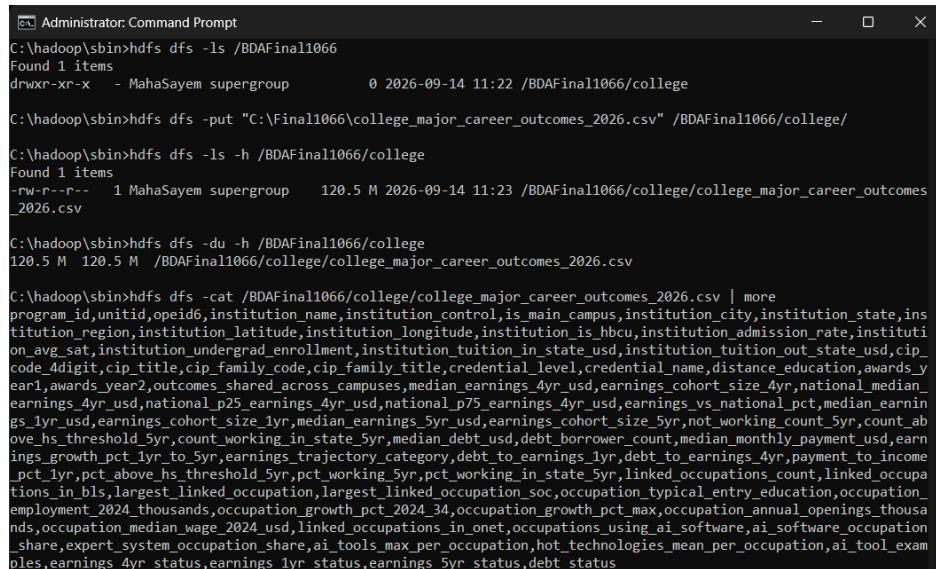
4.2 Uploading Dataset into HDFS

The dataset was uploaded from the local system using:

```
hdfs dfs -put "C:\Final1066\college_major_career_outcomes_2026.csv" /BDAFinal1066/college/
```

The uploaded file was verified using: `hdfs dfs -ls -h /BDAFinal1066/college`

The dataset was successfully stored with an HDFS size of 120.5 MB.



```
Administrator: Command Prompt
C:\hadoop\sbin>hdfs dfs -ls /BDAFinal1066
Found 1 items
drwxr-xr-x - MahaSayem supergroup 0 2026-09-14 11:22 /BDAFinal1066/college

C:\hadoop\sbin>hdfs dfs -put "C:\Final1066\college_major_career_outcomes_2026.csv" /BDAFinal1066/college/

C:\hadoop\sbin>hdfs dfs -ls -h /BDAFinal1066/college
Found 1 items
-rw-r--r-- 1 MahaSayem supergroup 120.5 M 2026-09-14 11:23 /BDAFinal1066/college/college_major_career_outcomes_2026.csv

C:\hadoop\sbin>hdfs dfs -du -h /BDAFinal1066/college
120.5 M 120.5 M /BDAFinal1066/college/college_major_career_outcomes_2026.csv

C:\hadoop\sbin>hdfs dfs -cat /BDAFinal1066/college/college_major_career_outcomes_2026.csv | more
program_id,unitid,opeid6,institution_name,institution_control,is_main_campus,institution_city,institution_state,institution_region,institution_latitude,institution_longitude,institution_is_hbcu,institution_admission_rate,institution_avg_sat,institution_undergrad_enrollment,institution_tuition_in_state_usd,institution_tuition_out_state_usd,cip_code_4digit,cip_title,cip_family_code,cip_family_title,credential_level,credential_name,distance_education,awards_year1,awards_year2,outcomes_shared_across_campuses,median_earnings_4yr_usd,earnings_cohort_size_4yr,national_median_earnings_4yr_usd,national_p25_earnings_4yr_usd,national_p75_earnings_4yr_usd,earnings_vs_national_pct,median_earnings_1yr_usd,earnings_cohort_size_1yr,median_earnings_5yr_usd,earnings_cohort_size_5yr,not_working_count_5yr,count_above_hs_threshold_5yr,count_working_in_state_5yr,median_debt_usd,debt_borrower_count,median_monthly_payment_usd,earnings_growth_pct_1yr_to_5yr,earnings_trajectory_category,debt_to_earnings_1yr,debt_to_earnings_4yr,payment_to_income_pct_1yr,pct_above_hs_threshold_5yr,pct_working_5yr,pct_working_in_state_5yr,linked_occupations_count,linked_occupations_in_bls,largest_linked_occupation,largest_linked_occupation_soc,occupation_typical_entry_education,occupation_employment_2024_thousands,occupation_growth_pct_2024_34,occupation_growth_pct_max,occupation_annual_openings_thousands,occupation_median_wage_2024_usd,linked_occupations_in_onet,occupations_using_ai_software,ai_software_occupation_share,expert_system_occupation_share,ai_tools_max_per_occupation,hot_technologies_mean_per_occupation,ai_tool_exampl
es,earnings_4yr_status,earnings_1yr_status,earnings_5yr_status,debt_status
```

Fig 2: Dataset uploaded successfully into HDFS

The screenshot shows the college_major_career_outcomes_2026.csv file successfully stored in the HDFS directory.

4.3 HDFS Block Analysis

The HDFS block size was checked using: `hdfs getconf -confKey dfs.blocksize`

The configured block size was: 134,217,728 bytes (128 MB)

The file's block information was then checked using:

```
hdfs fsck /BDAFinal1066/college/college_major_career_outcomes_2026.csv -files -blocks -locations
```

The analysis showed:

HDFS Property	Result
File Size	126,340,696 bytes
HDFS Blocks	1
Replication Factor	1
Missing Blocks	0
Corrupt Blocks	0
Under-replicated Blocks	0
HDFS Status	HEALTHY

```
Administrator: Command Prompt
C:\hadoop\sbin>hdfs getconf -confkey dfs.blocksize
134217728

C:\hadoop\sbin>hdfs fsck /BDAFinal1066/college/college_major_career_outcomes_2026.csv -files -blocks -locations
Connecting to namenode via http://localhost:9870/fsck?ugi=MahaSayem&files=1&blocks=1&locations=1&path=/BDAFinal1066/college/college_major_career_outcomes_2026.csv
FSCK started by MahaSayem (auth:SIMPLE) from /127.0.0.1 for path /BDAFinal1066/college/college_major_career_outcomes_2026.csv at Mon Sep 14 11:40:10 BDT 2026

/BDAFinal1066/college/college_major_career_outcomes_2026.csv 126340696 bytes, replicated: replication=1, 1 block(s): OK
0, BP-967097423-192.168.1.100-178751567811:blk_1073741983_1159 len=126340696 live_rep=1 [DataNodeInfoWithStorage[127.0.0.1:9866_DS-falaebc2-a20b-40e7-8fb7-e1b70e1d805c_DISK]]

Status: HEALTHY
Number of data-nodes: 1
Number of racks: 1
Total dirs: 0
Total symlinks: 0

Replicated blocks:
Total size: 126340696 B
Total files: 1
Total blocks (validated): 1 (avg. block size 126340696 B)
Minimally replicated blocks: 1 (100.0 %)
Over-replicated blocks: 0 (0.0 %)
Under-replicated blocks: 0 (0.0 %)
Mis-replicated blocks: 0 (0.0 %)
Default replication factor: 1
Average block replication: 1.0
Missing blocks: 0
Corrupt blocks: 0
Missing replicas: 0 (0.0 %)
Blocks queued for replication: 0

Erasure Coded Block Groups:
Total size: 0 B
Total files: 0
Total block groups (validated): 0
Minimally erasure-coded block groups: 0
Over-erasure-coded block groups: 0
Under-erasure-coded block groups: 0
Unsatisfactory placement block groups: 0
Average block group size: 0.0
Missing block groups: 0
Corrupt block groups: 0
Missing internal blocks: 0
Blocks queued for replication: 0
FSCK ended at Mon Sep 14 11:40:10 BDT 2026 in 3 milliseconds

The filesystem under path '/BDAFinal1066/college/college_major_career_outcomes_2026.csv' is HEALTHY
```

Fig 3: HDFS block and file system analysis

The fsck output confirms that the dataset occupies one HDFS block and the filesystem is HEALTHY, with no missing or corrupt blocks.

5. MapReduce Analysis

Java MapReduce was used to process the College Major & Career Outcomes 2026 dataset and perform three analytical operations. Each operation uses a Mapper to extract relevant fields and a Reducer to calculate the required result.

5.1 Average 4-Year Earnings by Major

Objective: The objective of this operation is to calculate the average 4-year earnings for each college major using Java MapReduce. This analysis comprises of:

1. **Driver Class:** Sets up the MapReduce job configuration, mapper and reducer classes, input/output paths, and output data types.
2. **Mapper Class:** Reads each CSV record and extracts the major code and 4-year median earnings. It emits the major code (Text) as the key and earnings (DoubleWritable) as the value.
3. **Reducer Class:** Groups the earnings values for each major and calculates the average earnings.

Code: CollegeMajorAverageEarnings.java

```
import java.io.IOException;
import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.DoubleWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.Mapper;
import org.apache.hadoop.mapreduce.Reducer;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
```

```

import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class CollegeMajorAverageEarnings {

    public static class EarningsMapper
        extends Mapper<Object, Text, Text, DoubleWritable> {

        private Text major = new Text();
        private DoubleWritable earnings = new DoubleWritable();

        public void map(Object key, Text value, Context context)
            throws IOException, InterruptedException {

            String line = value.toString();

            // Skip CSV header
            if (line.startsWith("program_id,")) {
                return;
            }

            String[] fields = parseCSV(line);

            // cip_title = column 18
            // median_earnings_4yr_usd = column 27
            if (fields.length > 27) {
                String majorName = fields[17].trim();
                String earningsValue = fields[27].trim();

                if (!majorName.isEmpty() && !earningsValue.isEmpty()) {
                    try {
                        double value4yr = Double.parseDouble(earningsValue);

                        major.set(majorName);
                        earnings.set(value4yr);

                        context.write(major, earnings);
                    } catch (NumberFormatException e) {
                        // Ignore invalid numeric values
                    }
                }
            }
        }
    }

    public static void main(String[] args) throws Exception {

        Configuration conf = new Configuration();

        Job job = Job.getInstance(conf, "Average 4 Year Earnings by Major");

        job.setJarByClass(CollegeMajorAverageEarnings.class);

        job.setMapperClass(EarningsMapper.class);
        job.setReducerClass(EarningsReducer.class);

        job.setOutputKeyClass(Text.class);
        job.setOutputValueClass(DoubleWritable.class);

        FileInputFormat.addInputPath(

```



```

        job, new Path(args[0]));

FileOutputFormat.setOutputPath(
    job, new Path(args[1]));

System.exit(job.waitForCompletion(true) ? 0 : 1);
    }
}

```

Output Snapshot:

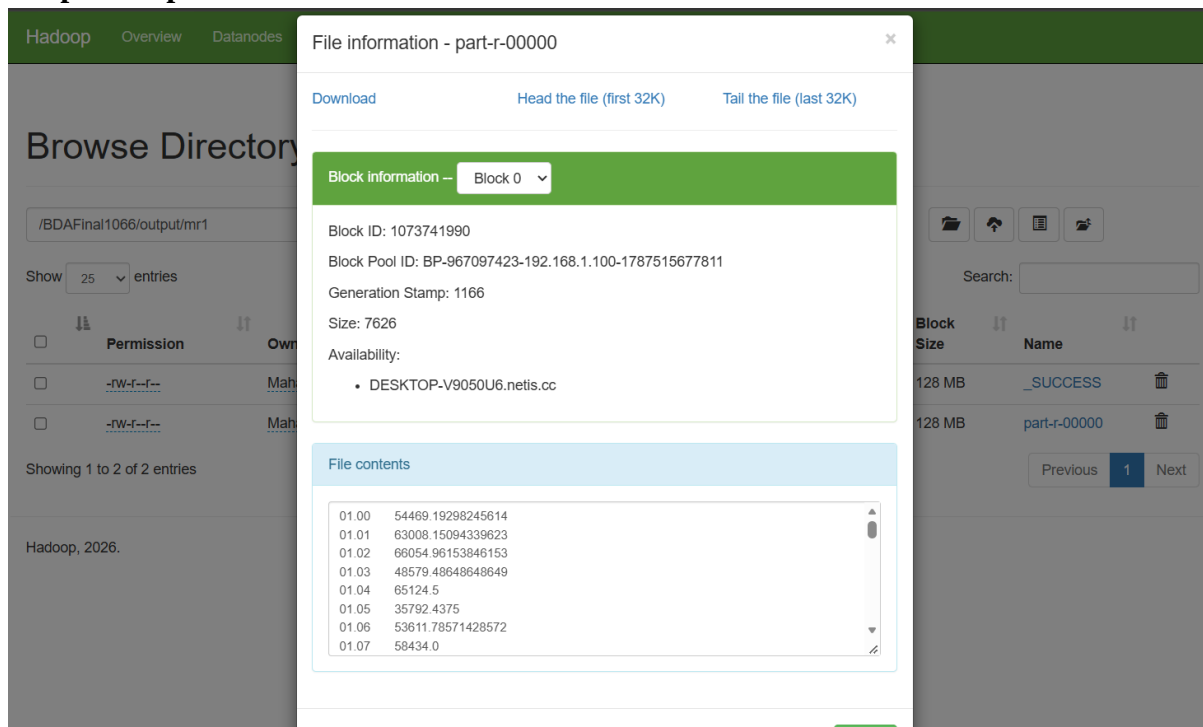


Fig 4: Average 4-Year Earnings by Major

5.2 Total Employment by Major Category

Objective: The objective of this operation is to calculate the total employment for each major category using Java MapReduce. This analysis comprises of:

1. **Driver Class:** Sets up the MapReduce job configuration, mapper and reducer classes, input/output paths, and output data types.
2. **Mapper Class:** Extracts the major category and employment value from each CSV record and emits the category (Text) as the key and employment (DoubleWritable) as the value.
3. **Reducer Class:** Groups the employment values by category and calculates the total employment.

Code: CollegeCategoryEmployment.java

```

import java.io.IOException;
import java.util.ArrayList;
import java.util.List;

import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;

```

```

import org.apache.hadoop.io.DoubleWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.Mapper;
import org.apache.hadoop.mapreduce.Reducer;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class CollegeCategoryEmployment {

    public static class EmploymentMapper
        extends Mapper<Object, Text, Text, DoubleWritable> {

        private Text category = new Text();
        private DoubleWritable employment = new DoubleWritable();

        public void map(Object key, Text value, Context context)
            throws IOException, InterruptedException {

            String line = value.toString();

            // Skip CSV header
            if (line.startsWith("program_id,")) {
                return;
            }

            String[] fields = parseCSV(line);

            /*
             * cip_family_title = column 21 → index 20
             * count_working_in_state_5yr = column 43 → index 42
             */
        }
    }

    public static class EmploymentReducer
        extends Reducer<Text, DoubleWritable, Text, DoubleWritable> {

        private DoubleWritable result = new DoubleWritable();

        public void reduce(Text key,
            Iterable<DoubleWritable> values,
            Context context)
            throws IOException, InterruptedException {

            double total = 0.0;

            for (DoubleWritable value : values) {
                total += value.get();
            }

            result.set(total);

            context.write(key, result);
        }

        public static void main(String[] args) throws Exception {
            Configuration conf = new Configuration();
            Job job = Job.getInstance(
                conf,
                "Total Employment by Major Category"
            );
        }
    }
}

```

```

);
job.setJarByClass(CollegeCategoryEmployment.class);
job.setMapperClass(EmploymentMapper.class);
job.setReducerClass(EmploymentReducer.class);
job.setOutputKeyClass(Text.class);
job.setOutputValueClass(DoubleWritable.class);

FileInputFormat.addInputPath(
    job,
    new Path(args[0])
);
FileOutputFormat.setOutputPath(
    job,
    new Path(args[1])
);
System.exit(
    job.waitForCompletion(true) ? 0 : 1
);
}
}

```

Output Snapshot:

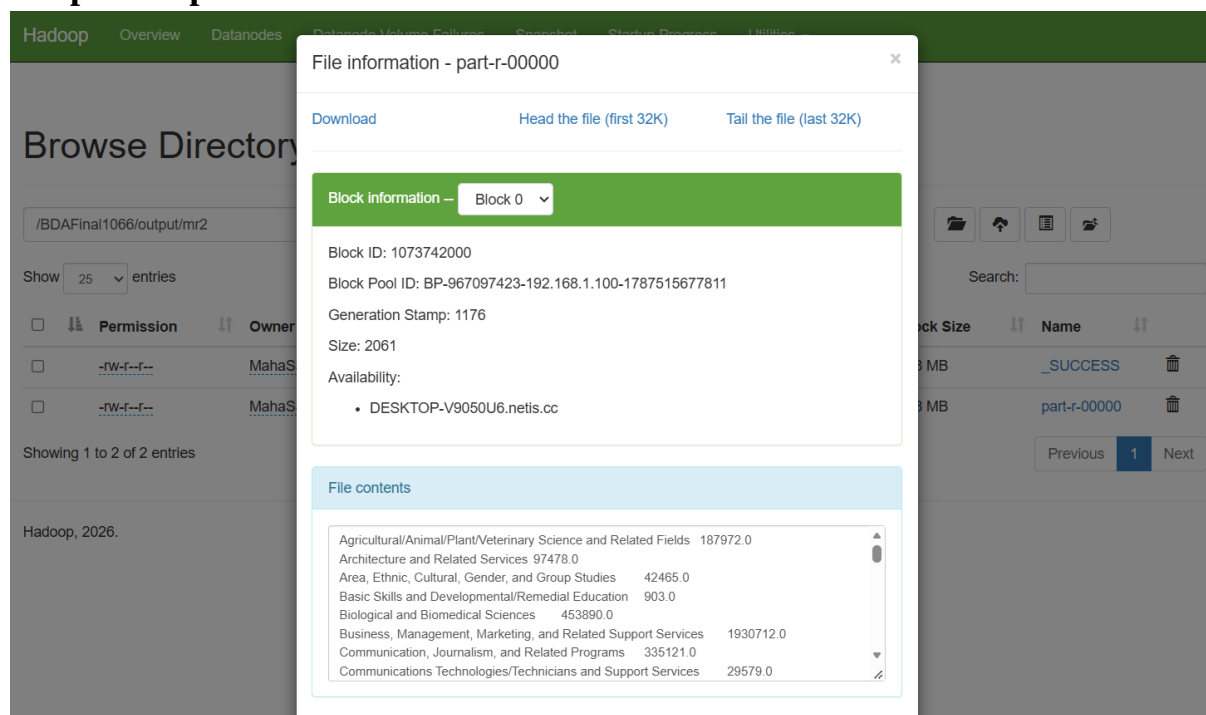


Fig 5: MapReduce Operation 2 – Total Employment by Major Category

5.3 Average Student Debt by Major Category

Objective: The objective of this operation is to calculate the average student debt for each major category using Java MapReduce. This analysis comprises of:

1. **Driver Class:** Sets up the MapReduce job configuration, mapper and reducer classes, input/output paths, and output data types.
2. **Mapper Class:** Extracts the major category and debt value from each CSV record and emits the category (Text) as the key and debt (DoubleWritable) as the value.

3. Reducer Class: Groups the debt values by category and calculates the average student debt.

Code: CollegeCategoryAverageDebt.java

```
import java.io.IOException;
import java.util.ArrayList;
import java.util.List;

import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.DoubleWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.Mapper;
import org.apache.hadoop.mapreduce.Reducer;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class CollegeCategoryAverageDebt {

    public static class DebtMapper
        extends Mapper<Object, Text, Text, DoubleWritable> {

        private Text category = new Text();
        private DoubleWritable debt = new DoubleWritable();

        public void map(Object key, Text value, Context context)
            throws IOException, InterruptedException {

            String line = value.toString();

            // Skip CSV header
            if (line.startsWith("program_id,")) {
                return;
            }

            String[] fields = parseCSV(line);

            /*
             * cip_family_title = column 21 → index 20
             * median_debt_usd = column 40 → index 39
             */

            if (fields.length > 39) {

                String categoryName = fields[20].trim();
                String debtValue = fields[39].trim();

                if (!categoryName.isEmpty() && !debtValue.isEmpty()) {

                    try {

                        double debtAmount =
                            Double.parseDouble(debtValue);

                        category.set(categoryName);
                        debt.set(debtAmount);
```

```

        context.write(category, debt);

    } catch (NumberFormatException e) {
        // Ignore invalid or suppressed values
    }
}
}

public static class DebtReducer
    extends Reducer<Text, DoubleWritable, Text, DoubleWritable> {

    private DoubleWritable result = new DoubleWritable();

    public void reduce(Text key,
        Iterable<DoubleWritable> values,
        Context context)
        throws IOException, InterruptedException {

        double sum = 0.0;
        long count = 0;

        for (DoubleWritable value : values) {

            sum += value.get();
            count++;
        }

        if (count > 0) {

            result.set(sum / count);

            context.write(key, result);
        }
    }
}

public static void main(String[] args) throws Exception {

    Configuration conf = new Configuration();

    Job job = Job.getInstance(
        conf,
        "Average Student Debt by Major Category"
    );

    job.setJarByClass(CollegeCategoryAverageDebt.class);

    job.setMapperClass(DebtMapper.class);
    job.setReducerClass(DebtReducer.class);

    job.setOutputKeyClass(Text.class);
    job.setOutputValueClass(DoubleWritable.class);

    FileInputFormat.addInputPath(
        job,
        new Path(args[0])
    );
}

```

```

        FileOutputFormat.setOutputPath(
            job,
            new Path(args[1])
        );

        System.exit(
            job.waitForCompletion(true) ? 0 : 1
        );
    }
}

```

Output Snapshot:

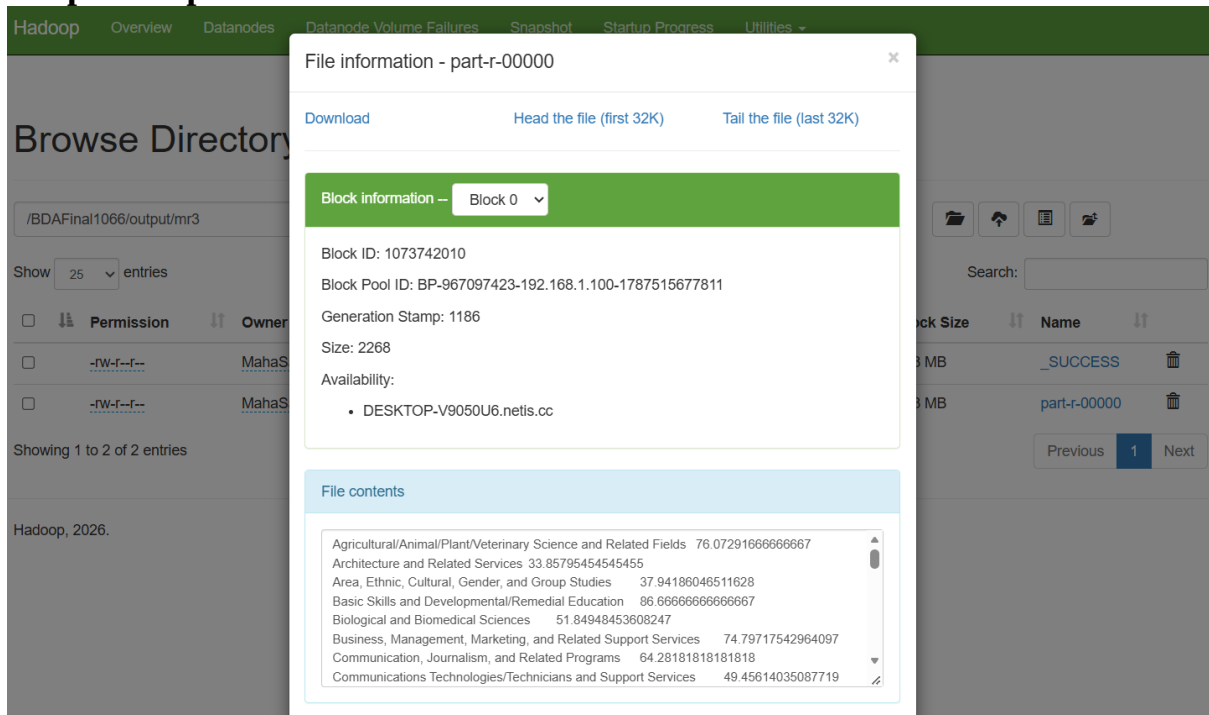


Fig 6: MapReduce Operation 3 – Average Student Debt by Major Category

6. Pig Analysis

6.1 High-Earning Programs

Objective: The objective of this operation is to identify college programs with 4-year median earnings above \$70,000 using Apache Pig. This analysis comprises of:

1. **LOAD:** Loads the dataset from HDFS using CSVExcelStorage().
2. **FILTER:** Selects records where median_earnings_4yr_usd > 70000.
3. **FOREACH:** Extracts the major title and 4-year median earnings.
4. **STORE:** Stores the filtered results in HDFS.

Pig Script: pig1_high_earning_programs.pig

```

data = LOAD '/BDAFinal1066/college/college_major_career_outcomes_2026.csv'
USING PigStorage(',')
AS (
    program_id:chararray,
    unitid:chararray,

```

```

opeid6:chararray,
institution_name:chararray,
institution_control:chararray,
is_main_campus:int,
institution_city:chararray,
institution_state:chararray,
institution_region:chararray,
institution_latitude:double,
institution_longitude:double,
institution_is_hbcu:int,
institution_admission_rate:double,
institution_avg_sat:double,
institution_undergrad_enrollment:int,
institution_tuition_in_state_usd:double,
institution_tuition_out_state_usd:double,
cip_code_4digit:chararray,
cip_title:chararray,
cip_family_code:chararray,
cip_family_title:chararray,
credential_level:int,
credential_name:chararray,
distance_education:chararray,
awards_year1:double,
awards_year2:double,
outcomes_shared_across_campuses:int,
median_earnings_4yr_usd:double
);
filtered_data = FILTER data BY median_earnings_4yr_usd > 70000;
result = FOREACH filtered_data GENERATE
    cip_title,
    median_earnings_4yr_usd;
STORE result INTO '/BDAFinal1066/output/pig1';

```

Output Snapshot:

The screenshot displays the Hadoop web interface. A modal window titled 'File information - part-m-00000' is open, showing details for a file in the HDFS. The file is located at '/BDAFinal1066/output/pig1'. The modal includes a 'Block information' section with the following details:

- Block ID: 1073742023
- Block Pool ID: BP-967097423-192.168.1.100-1787515677811
- Generation Stamp: 1199
- Size: 193205
- Availability:
 - DESKTOP-V9050U6.netis.cc

Below the block information, the 'File contents' section shows a list of programs and their median earnings:

- Electromechanical Technologies/Technicians 70658.0
- Electromechanical Technologies/Technicians 92611.0
- Medicine 130097.0
- Medicine 176093.0
- Medicine 118072.0
- International Relations and National Security Studies 87991.0
- International Relations and National Security Studies 86451.0
- Computer/Information Technology Administration and Management 72488.0

The background shows the Hadoop 'Browse Directory' page with a table of files. The table has columns for 'Permission', 'Owner', and 'Name'. The file 'part-m-00000' is visible in the table.

Fig 7: Pig Operation 1 – High-Earning Programs

6.2 Average Earnings by Major Category

Objective: The objective of this operation is to calculate the average 4-year earnings for each major category using Apache Pig. This analysis comprises of:

1. **LOAD:** Loads the dataset from HDFS using `CSVExcelStorage()`.
2. **FILTER:** Removes records with missing major categories or earnings values.
3. **GROUP:** Groups the valid records by `cip_family_title`.
4. **FOREACH:** Calculates the average 4-year earnings for each major category.
5. **STORE:** Stores the results in HDFS.

Pig Script: `pig2_average_earnings_by_category.pig`

```
REGISTER 'file:///C:/pig/lib/piggybank.jar';
```

```
data = LOAD '/BDAFinal1066/college/college_major_career_outcomes_2026.csv'
USING org.apache.pig.piggybank.storage.CSVExcelStorage()
```

```
AS (
    program_id:chararray,
    unitid:chararray,
    opeid6:chararray,
    institution_name:chararray,
    institution_control:chararray,
    is_main_campus:int,
    institution_city:chararray,
    institution_state:chararray,
    institution_region:chararray,
    institution_latitude:double,
    institution_longitude:double,
    institution_is_hbcu:int,
    institution_admission_rate:double,
    institution_avg_sat:double,
    institution_undergrad_enrollment:int,
    institution_tuition_in_state_usd:double,
    institution_tuition_out_state_usd:double,
    cip_code_4digit:chararray,
    cip_title:chararray,
    cip_family_code:chararray,
    cip_family_title:chararray,
    credential_level:int,
    credential_name:chararray,
    distance_education:chararray,
    awards_year1:double,
    awards_year2:double,
    outcomes_shared_across_campuses:int,
    median_earnings_4yr_usd:double
);
valid_data = FILTER data BY
    cip_family_title IS NOT NULL
    AND median_earnings_4yr_usd IS NOT NULL;
grouped_data = GROUP valid_data BY cip_family_title;
result = FOREACH grouped_data GENERATE
    group AS major_category,
    AVG(valid_data.median_earnings_4yr_usd) AS average_earnings;
STORE result INTO '/BDAFinal1066/output/pig2_corrected';
```


Output Snapshot:

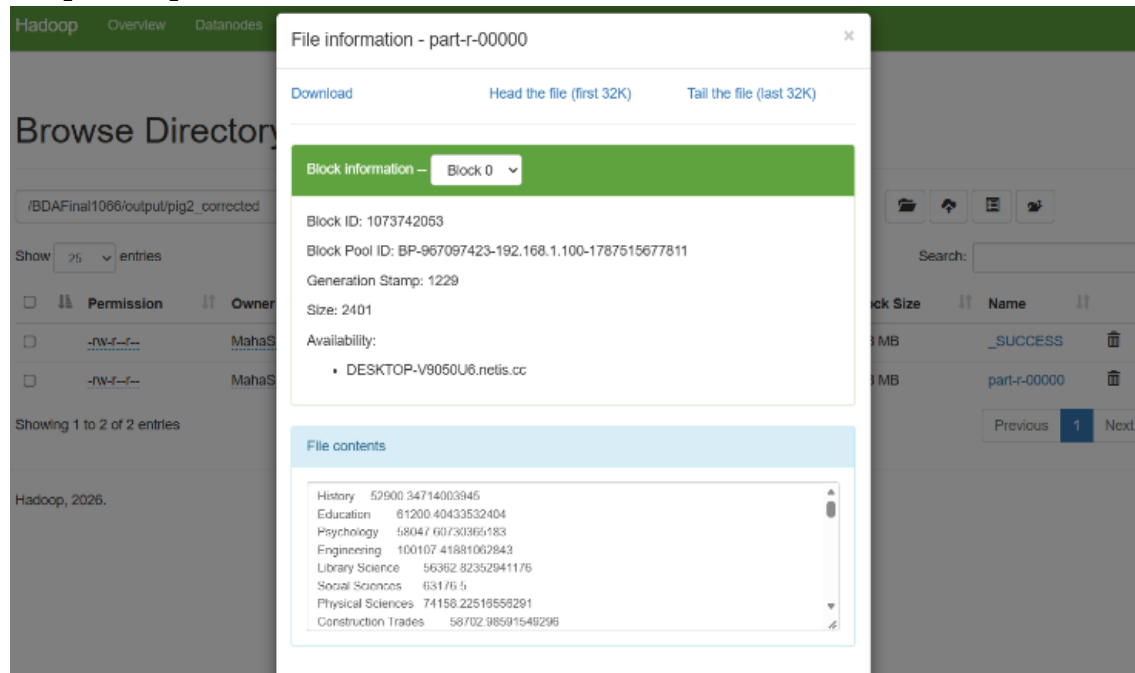


Fig 8: Pig Operation 2 – Average Earnings by Major Category

6.3 Total Employment by Major Category

Objective: The objective of this operation is to calculate the total employment for each major category using Apache Pig. This analysis comprises of:

1. **LOAD:** Loads the dataset from HDFS using `CSVExcelStorage()`.
2. **FILTER:** Removes records with missing category or employment values.
3. **GROUP:** Groups the records by `cip_family_title`.
4. **FOREACH:** Calculates the total employment using `SUM()`.
5. **STORE:** Stores the results in HDFS.

Pig Script: `pig3_total_employment_by_category.pig`

```
REGISTER 'file:///C:/pig/lib/piggybank.jar';
```

```
data = LOAD '/BDAFinal1066/college/college_major_career_outcomes_2026.csv'
USING org.apache.pig.piggybank.storage.CSVExcelStorage()
AS (
    program_id:chararray,
    unitid:chararray,
    opeid6:chararray,
    institution_name:chararray,
    institution_control:chararray,
    is_main_campus:int,
    institution_city:chararray,
    institution_state:chararray,
    institution_region:chararray,
    institution_latitude:double,
    institution_longitude:double,
    institution_is_hbcu:int,
    institution_admission_rate:double,
    institution_avg_sat:double,
```

```

institution_undergrad_enrollment:int,
institution_tuition_in_state_usd:double,
institution_tuition_out_state_usd:double,
cip_code_4digit:chararray,
cip_title:chararray,
cip_family_code:chararray,
cip_family_title:chararray,
credential_level:int,
credential_name:chararray,
distance_education:chararray,
awards_year1:double,
awards_year2:double,
outcomes_shared_across_campuses:int,
median_earnings_4yr_usd:double,
earnings_cohort_size_4yr:double,
national_median_earnings_4yr_usd:double,
national_p25_earnings_4yr_usd:double,
national_p75_earnings_4yr_usd:double,
earnings_vs_national_pct:double,
median_earnings_1yr_usd:double,
earnings_cohort_size_1yr:double,
median_earnings_5yr_usd:double,
earnings_cohort_size_5yr:double,
not_working_count_5yr:double,
count_above_hs_threshold_5yr:double,
count_working_in_state_5yr:double
);
valid_data = FILTER data BY
    cip_family_title IS NOT NULL
    AND count_working_in_state_5yr IS NOT NULL;
grouped_data = GROUP valid_data BY cip_family_title;
result = FOREACH grouped_data GENERATE
    group AS major_category,
    SUM(valid_data.count_working_in_state_5yr) AS total_employment;
STORE result INTO '/BDAFinal1066/output/pig3';

```

Output Snapshot:

The screenshot shows the Hadoop web interface. A modal window titled 'File information - part-r-00000' is open, displaying details for a file in the HDFS. The file is located at '/BDAFinal1066/output/pig3'. The modal shows block information for Block 0, including its ID, pool ID, generation stamp, size, and availability. Below this, the file contents are displayed as a table with major categories and their corresponding counts.

Block information	Block 0
Block ID:	1073742068
Block Pool ID:	BP-967097423-192.168.1.100-1787515677811
Generation Stamp:	1244
Size:	1871
Availability:	DESKTOP-V9050U6.netis.cc

File contents	
History	13158.0
Education	154877.0
Psychology	102985.0
Engineering	57350.0
Library Science	2706.0
Social Sciences	72233.0
Physical Sciences	6995.0
Construction Trades	15441.0

Fig 9: Pig Operation 3 – Total Employment by Major Category

6.4 Top 10 Employment Categories

Objective: The objective of this operation is to identify the top 10 major categories based on total employment using Apache Pig. This analysis comprises of:

1. **LOAD:** Loads the dataset from HDFS using `CSVExcelStorage()`.
2. **FILTER:** Removes records with missing category or employment values.
3. **GROUP:** Groups the records by major category.
4. **FOREACH:** Calculates total employment for each category using `SUM()`.
5. **ORDER:** Sorts the categories by total employment in descending order.
6. **LIMIT:** Selects the top 10 categories.
7. **STORE:** Stores the results in HDFS.

Pig Script: `pig4_top10_employment.pig`

```
REGISTER 'file:///C:/pig/lib/piggybank.jar';
```

```
data = LOAD '/BDAFinal1066/college/college_major_career_outcomes_2026.csv'
USING org.apache.pig.piggybank.storage.CSVExcelStorage()
```

```
AS (
  program_id:chararray,
  unitid:chararray,
  opeid6:chararray,
  institution_name:chararray,
  institution_control:chararray,
  is_main_campus:int,
  institution_city:chararray,
  institution_state:chararray,
  institution_region:chararray,
  institution_latitude:double,
  institution_longitude:double,
  institution_is_hbcu:int,
  institution_admission_rate:double,
  institution_avg_sat:double,
  institution_undergrad_enrollment:int,
  institution_tuition_in_state_usd:double,
  institution_tuition_out_state_usd:double,
  cip_code_4digit:chararray,
  cip_title:chararray,
  cip_family_code:chararray,
  cip_family_title:chararray,
  credential_level:int,
  credential_name:chararray,
  distance_education:chararray,
  awards_year1:double,
  awards_year2:double,
  outcomes_shared_across_campuses:int,
  median_earnings_4yr_usd:double,
  earnings_cohort_size_4yr:double,
  national_median_earnings_4yr_usd:double,
  national_p25_earnings_4yr_usd:double,
  national_p75_earnings_4yr_usd:double,
  earnings_vs_national_pct:double,
  median_earnings_1yr_usd:double,
  earnings_cohort_size_1yr:double,
  median_earnings_5yr_usd:double,
  earnings_cohort_size_5yr:double,
```

```

        not_working_count_5yr:double,
        count_above_hs_threshold_5yr:double,
        count_working_in_state_5yr:double
    );

valid_data = FILTER data BY
    cip_family_title IS NOT NULL
    AND count_working_in_state_5yr IS NOT NULL;

grouped_data = GROUP valid_data BY cip_family_title;

employment = FOREACH grouped_data GENERATE
    group AS major_category,
    SUM(valid_data.count_working_in_state_5yr) AS total_employment;

ordered = ORDER employment BY total_employment DESC;

top10 = LIMIT ordered 10;

STORE top10 INTO '/BDAFinal1066/output/pig4';

```

Output Snapshot:

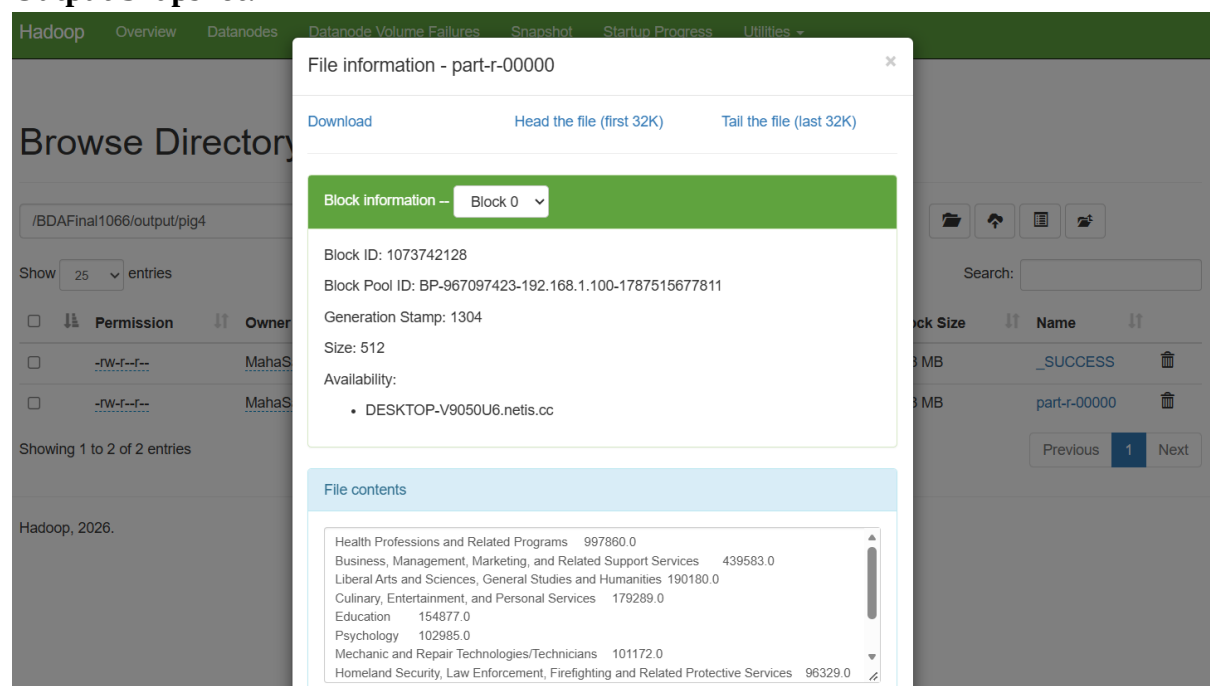


Fig 10: Pig Operation 4 – Top 10 Employment Categories

6.5 High Earnings and Low Debt Programs

Objective: The objective of this operation is to identify college programs with 4-year median earnings above \$70,000 and median debt below \$30,000 using Apache Pig. This analysis comprises of:

1. **LOAD:** Loads the dataset from HDFS using `CSVExcelStorage()`.
2. **FILTER:** Selects programs with earnings above \$70,000 and debt below \$30,000.
3. **FOREACH:** Extracts the major, earnings, and debt values.
4. **STORE:** Stores the filtered results in HDFS.

```

Pig Script: pig5_high_earning_low_debt.pig
REGISTER 'file:///C:/pig/lib/piggybank.jar';

data = LOAD '/BDAFinal1066/college/college_major_career_outcomes_2026.csv'
USING org.apache.pig.piggybank.storage.CSVExcelStorage()
AS (
    program_id:chararray,
    unitid:chararray,
    opeid6:chararray,
    institution_name:chararray,
    institution_control:chararray,
    is_main_campus:int,
    institution_city:chararray,
    institution_state:chararray,
    institution_region:chararray,
    institution_latitude:double,
    institution_longitude:double,
    institution_is_hbcu:int,
    institution_admission_rate:double,
    institution_avg_sat:double,
    institution_undergrad_enrollment:int,
    institution_tuition_in_state_usd:double,
    institution_tuition_out_state_usd:double,
    cip_code_4digit:chararray,
    cip_title:chararray,
    cip_family_code:chararray,
    cip_family_title:chararray,
    credential_level:int,
    credential_name:chararray,
    distance_education:chararray,
    awards_year1:double,
    awards_year2:double,
    outcomes_shared_across_campuses:int,
    median_earnings_4yr_usd:double,
    earnings_cohort_size_4yr:double,
    national_median_earnings_4yr_usd:double,
    national_p25_earnings_4yr_usd:double,
    national_p75_earnings_4yr_usd:double,
    earnings_vs_national_pct:double,
    median_earnings_1yr_usd:double,
    earnings_cohort_size_1yr:double,
    median_earnings_5yr_usd:double,
    earnings_cohort_size_5yr:double,
    not_working_count_5yr:double,
    count_above_hs_threshold_5yr:double,
    count_working_in_state_5yr:double,
    median_debt_usd:double
);

filtered_data = FILTER data BY
    median_earnings_4yr_usd > 70000
    AND median_debt_usd < 30000;

result = FOREACH filtered_data GENERATE
    cip_title AS major,
    median_earnings_4yr_usd AS earnings,
    median_debt_usd AS debt;

STORE result INTO '/BDAFinal1066/output/pig5';

```

Output Snapshot:

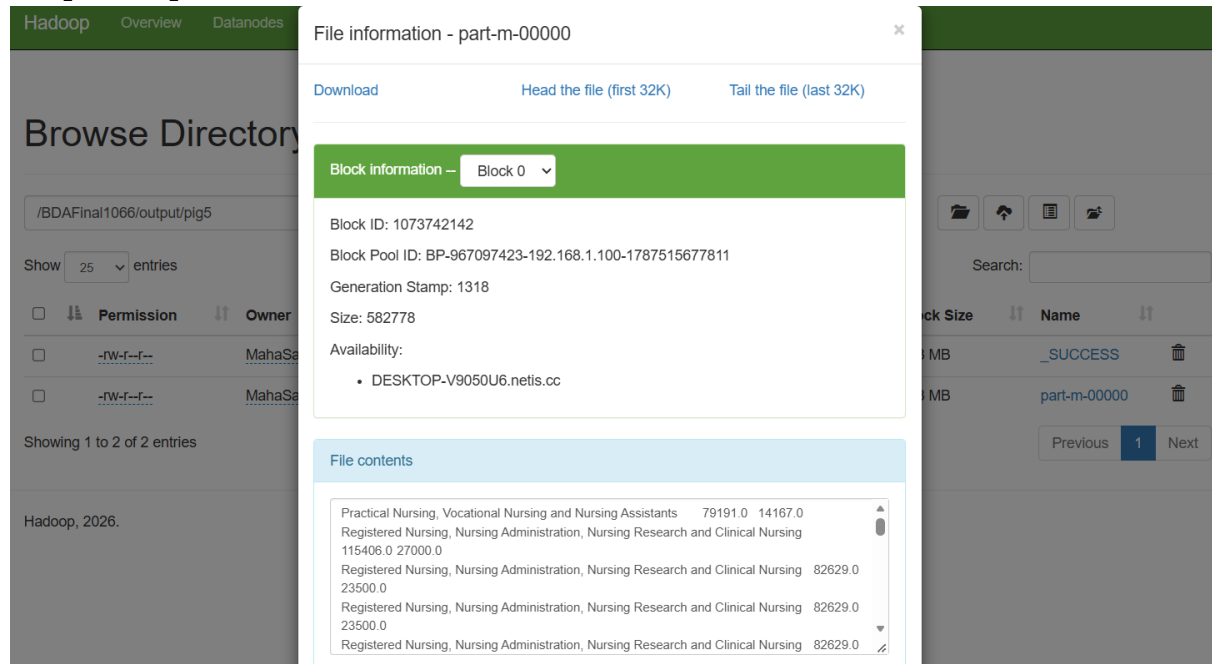


Fig 11: Pig Operation 5 – High Earnings and Low Debt Programs

7. Result

The project successfully completed the analysis using HDFS, Java MapReduce, and Apache Pig. The following tables present representative output values obtained from each operation.

7.1 MapReduce Results

ID	Analysis	Sample Output
MR1	Average 4-Year Earnings by Major	01.00 → 54,469.19 01.01 → 63,008.15 01.02 → 66,054.96
MR2	Total Employment by Category	Agricultural → 187,972 Architecture → 97,478 Business → 1,930,712
MR3	Average Student Debt by Category	Agricultural → 76.07 Architecture → 33.86 Area Studies → 37.94

7.2 Apache Pig Results

ID	Analysis	Sample Output
PIG1	High-Earning Programs	Medicine → \$130,097 International Relations → \$87,991 Electromechanical Technologies → \$70,658
PIG2	Average Earnings by Category	Engineering → \$100,107.42

		Legal Professions → \$83,061.48 Business → \$73,390.33
PIG3	Total Employment by Category	Health Professions → 997,860 Business → 439,583 Liberal Arts → 190,180
PIG4	Top 10 Employment Categories	Health Professions → 997,860 Business → 439,583 Liberal Arts → 190,180
PIG5	High Earnings + Low Debt	Practical Nursing → \$79,191 / \$14,167 Registered Nursing → \$115,406 / \$27,000 Registered Nursing → \$82,629 / \$23,500

8. Conclusion

This project gave us practical experience in working with a real-world college career outcomes dataset using Hadoop HDFS, MapReduce, and Apache Pig. We first stored and verified the dataset in HDFS, then used three MapReduce operations to analyze earnings, employment, and student debt. After that, five Pig operations were performed to filter, group, aggregate, sort, and identify useful patterns in the data. The analysis showed some interesting results. Health Professions and Related Programs had the highest total employment, while Engineering showed a comparatively high average earnings value. The final Pig operation also helped identify programs that offer higher earnings with relatively lower student debt.

Overall, the project helped us understand how Big Data technologies can be used to handle a large dataset and turn raw information into meaningful results. It also provided hands-on experience with distributed storage, data processing, aggregation, filtering, and result analysis, which are important concepts in Big Data Analytics.

9. Appendix

Appendix A: Project Resources

- **GitHub Repository:** [BDAL Final Project](#)
- **Dataset:** College Major & Career Outcomes 2026 - sourced from Kaggle
- **HDFS Dataset Path:**
/BDAFinal1066/college/college_major_career_outcomes_2026.csv

Appendix B: Technologies Used

- Apache Hadoop & HDFS
- Java MapReduce
- Apache Pig
- Piggybank CSVExcelStorage()
- Git & GitHub